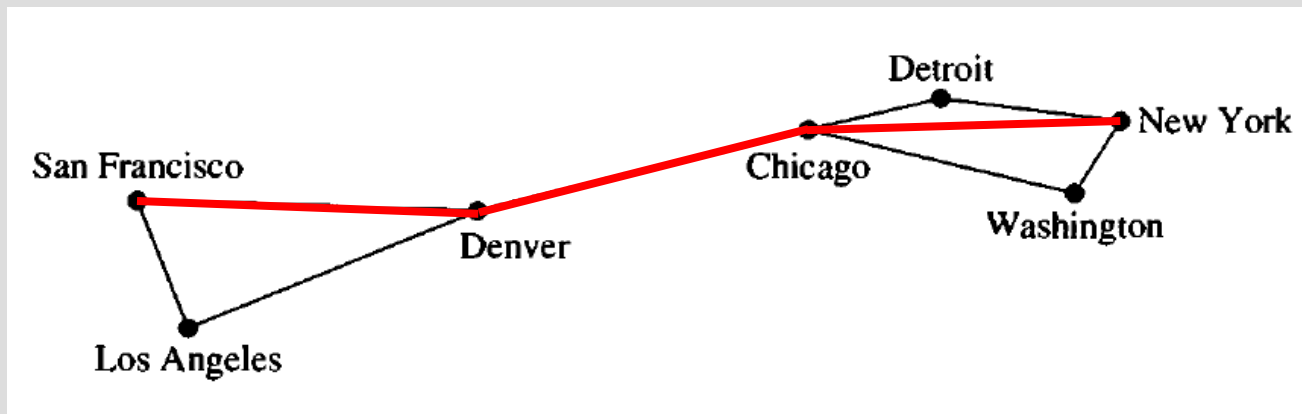


GRAPHS

CSI 219 – Discrete Mathematics

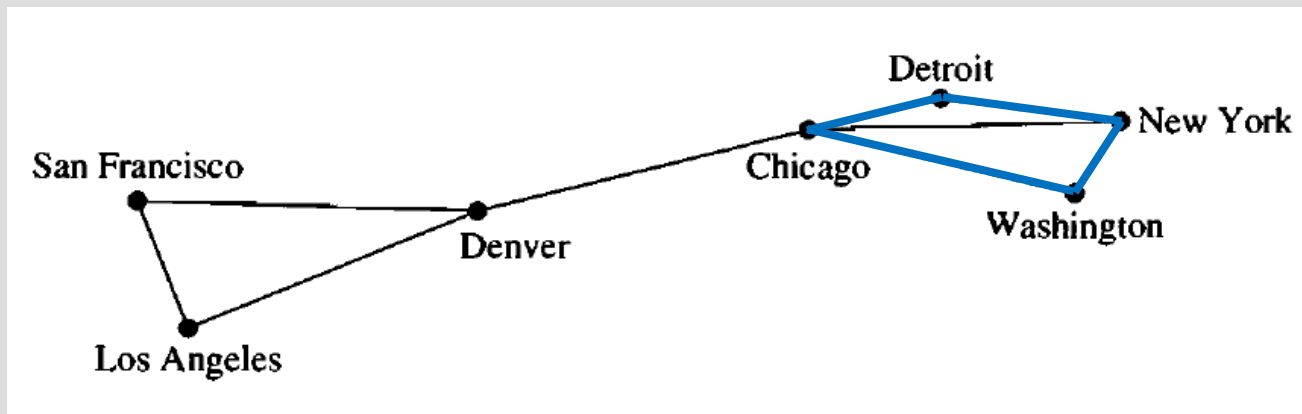
GRAPH CONNECTIVITY

PATH AND CIRCUIT



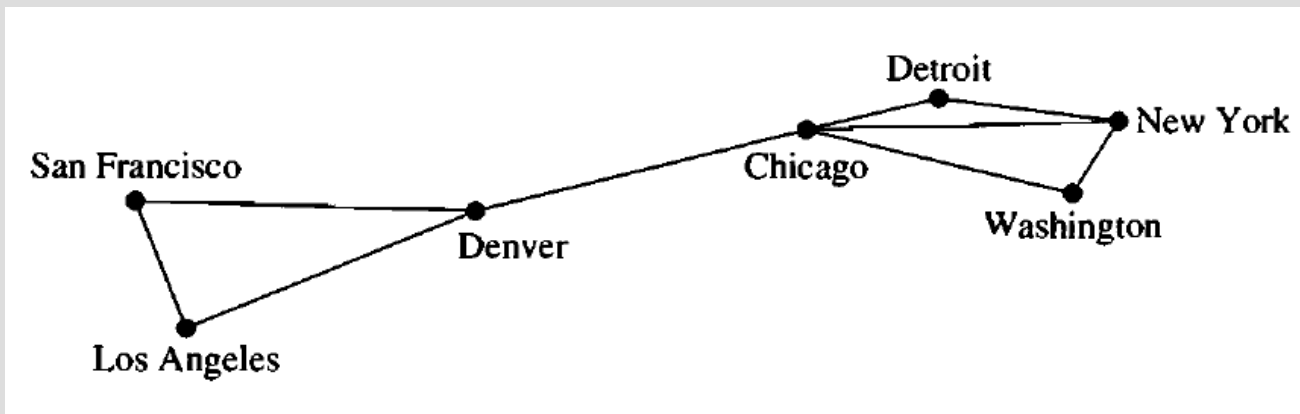
- **A path** is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph.
- There is a **path** from SF to NY
 - The path length is 3

PATH AND CIRCUIT



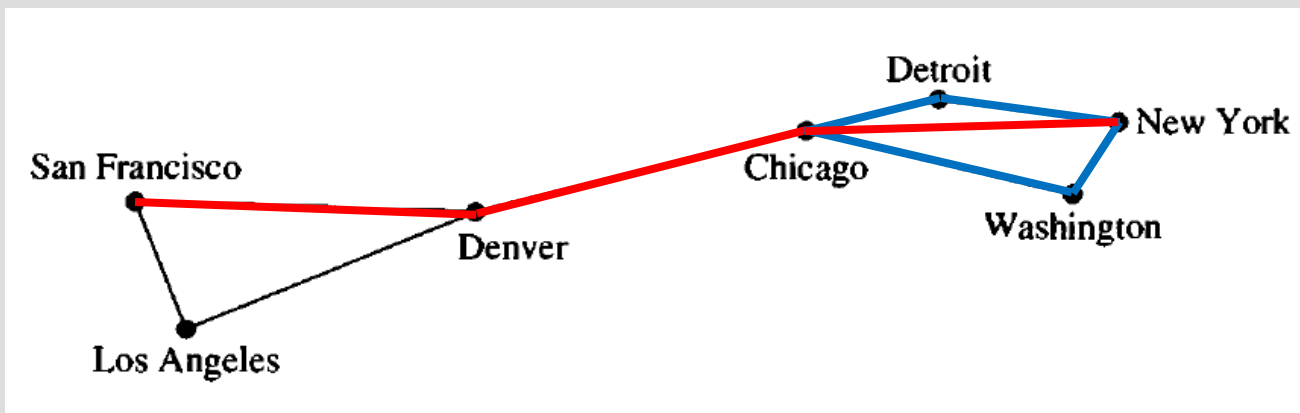
- The path is a circuit if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero.
- There is a **circuit** from Detroit to Detroit via NY, Washington and Chicago
 - The cycle length is 4

PATH AND CIRCUIT



- Is SF-Denver-Chicago-Denver-LA a path from SF to LA?
- Is Denver-Chicago-Denver a circuit from Denver to Chicago?

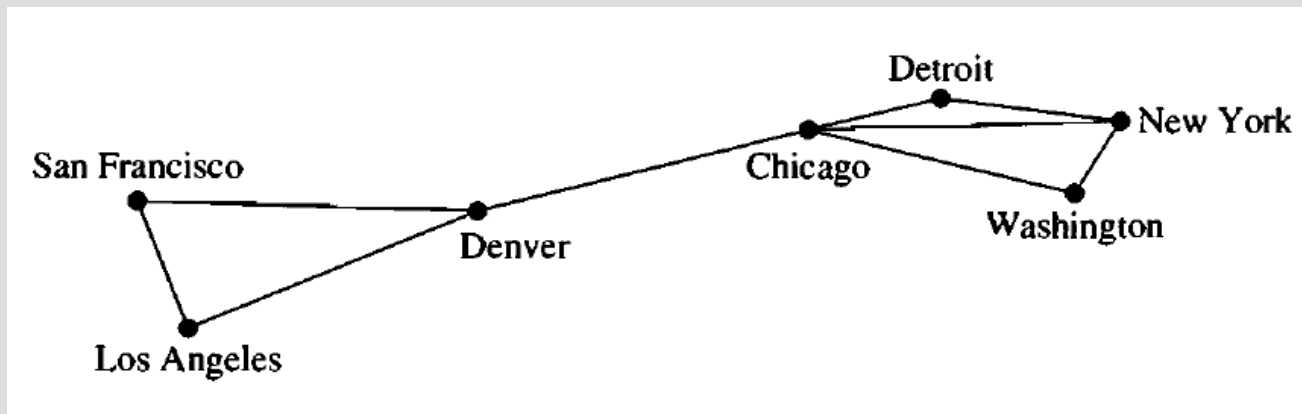
SIMPLE PATH AND SIMPLE CIRCUIT



- A path with no repeated edge is a **simple path**
- A circuit with no repeated edge is a **simple circuit**

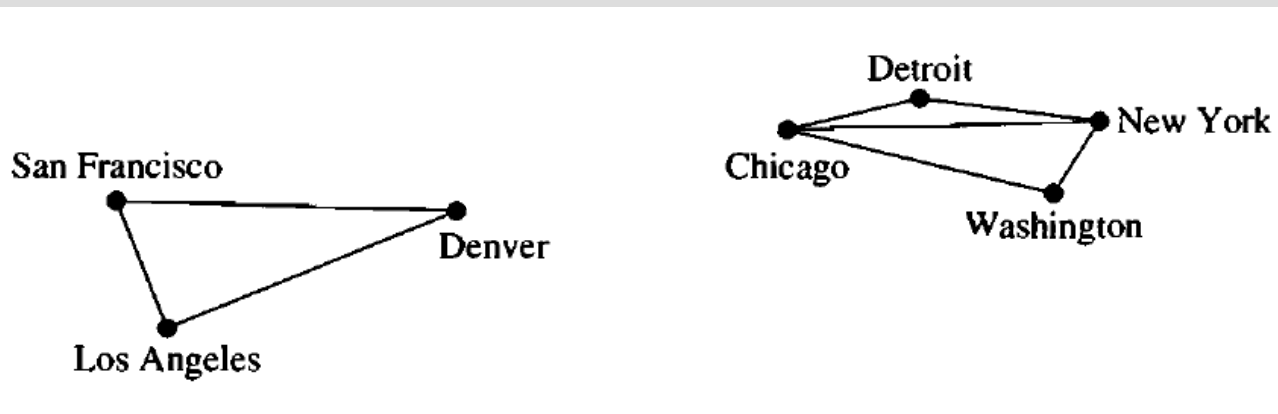
CONNECTEDNESS IN UNDIRECTED GRAPHS

CONNECTED GRAPH



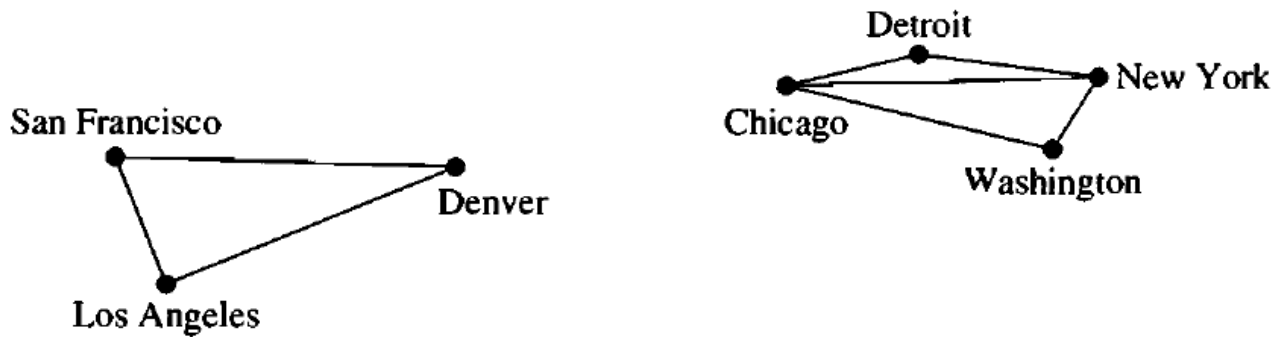
- An undirected graph is called **connected** if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not connected is called disconnected.
- We can reach any vertex from any other vertex in this graph
- This graph is a connected graph

CONNECTED GRAPH



- We cannot reach LA from NY
- So the graph is not connected

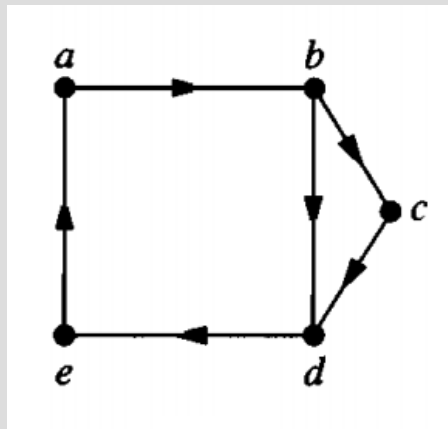
CONNECTED COMPONENTS



- A connected component of a graph G is a maximal connected subgraph of G .
- We can communicate between LA, SF and Denver
- We can communicate between Detroit, NY, Chicago and Washington
- These are connected components

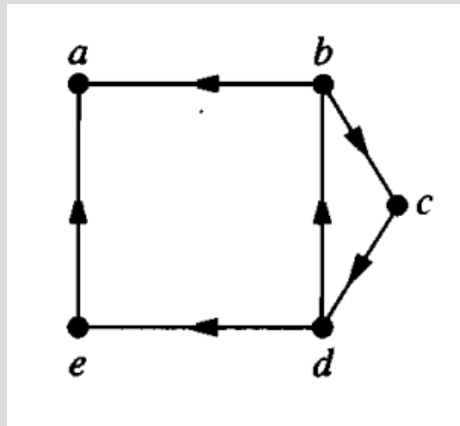
CONNECTEDNESS IN DIRECTED GRAPHS

STRONGLY CONNECTED GRAPH



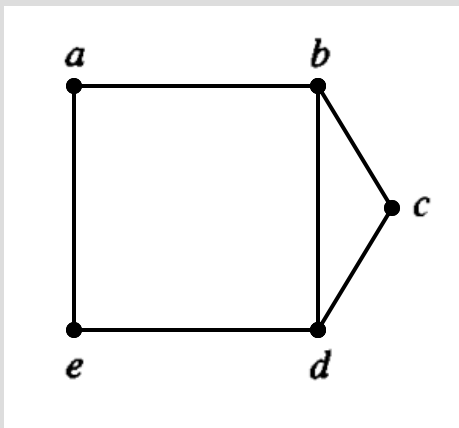
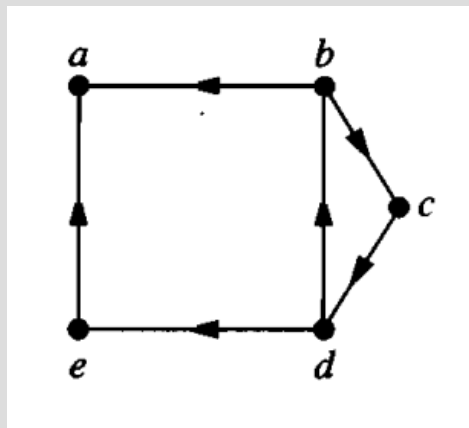
- A *directed graph* is strongly connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph.

WEAKLY CONNECTED GRAPH



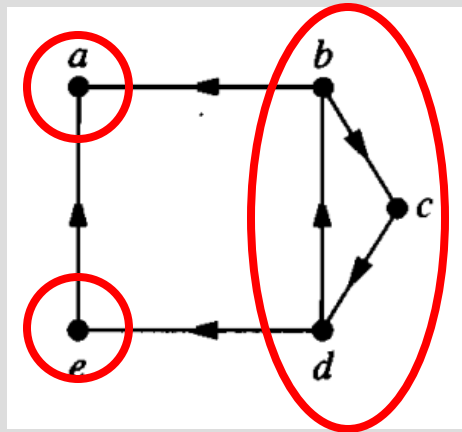
- We cannot reach from a to b
- So the graph is not strongly connected
- A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.

WEAKLY CONNECTED GRAPH



- We say a directed graph is weakly connected if its underlying undirected graph is connected.

STRONGLY CONNECTED COMPONENTS



- The subgraphs of a directed graph G that are strongly connected but not contained in larger strongly connected subgraphs, that is, the maximal strongly connected subgraphs, are called the *strongly connected components* or *strong components* of G .

EXERCISE

Find out strongly connected components of this graph:

